



## Chapter 4. Estimation

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# Population and sample

## Definition 4.1:

- A population is the set of all individuals of interest. The number of individuals  $N$  is called the population size.

# Population and sample

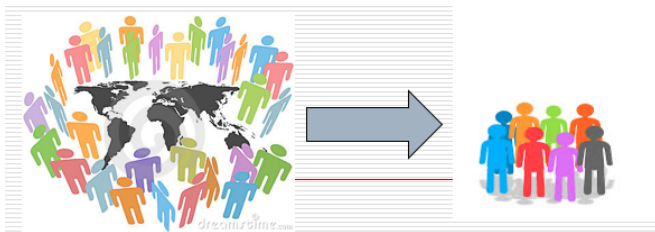
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- In practice, we usually study on a characteristic  $X$  of individuals in a population. Let  $x_i$  be the value of  $X$  of the  $i^{th}$  individual, the set  $\{x_1, x_2, \dots, x_N\}$  or  $X$  is also called the population.

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- A subset of  $n$  individuals taken from a population  $X$  is called a sample of size  $n$ . A sample of size  $n$  is a vector of  $n$  observations  $(x_1, x_2, \dots, x_n)$ .



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- A sample  $(x_1, x_2, \dots, x_n)$  is called a representation of random sample  $(X_1, X_2, \dots, X_n)$ .
- The joint probability distribution (pdf or pmf) of the random sample  $(X_1, X_2, \dots, X_n)$  is

$$f(x_1, x_2, \dots, x_n) = f(x_1)f(x_2) \dots f(x_n)$$

and is called the likelihood function.

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- A random vector  $(X_1, X_2, \dots, X_{50})$ , where  $X_i$  are i.i.d random variable having the same normal distribution  $f(x; \mu, \sigma^2)$ , is called a random sample of size 50 drawn from the population  $X$ .

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- Observed the electricity bills of 50 households from this region and obtained the following sample  $(x_1, x_2, \dots, x_{50}) = (255, 367, \dots, 423)$ , this sample is a representation of the random sample  $(X_1, X_2, \dots, X_{50})$ .

# Statistic

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- A statistic is a random variable and the distribution of a statistic is called a sampling distribution.

# Some important statistic

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- The adjusted random sample standard deviation  $S = \sqrt{S^2}$

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# Sampling distribution of random sample mean

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- Theorem 4.1:** For all distribution of  $X$ , we have  $E(\bar{X}) = \mu$  and  $V(\bar{X}) = \frac{\sigma^2}{n}$ .
- Theorem 4.2:** If  $X$  is normal:  $X \sim N(\mu, \sigma^2)$  then  $\bar{X}$  is also normal:  $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ . So the Z-statistic is standard normal:  
 $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$

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- Theorem 4.3:** The central limit theorem:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{n \rightarrow +\infty} N(0, 1)$$

When  $n$  is large enough,  $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1)$  or  $\bar{X} \approx N(\mu, \frac{\sigma^2}{n})$ , for all distribution of  $X$ .

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- Consider a random sample  $(X_1, X_2, \dots, X_n)$  taken from a population  $X$ . Denote by  $\mu = E(X)$  and  $\sigma^2 = V(X)$ .
- **Theorem 4.4:** If  $X$  is normal:  $X \sim N(\mu, \sigma^2)$  then the T-statistic follows the Student's distribution with  $n - 1$  degrees of freedom:

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- Theorem 4.5** (The central limit theorem + Slutsky's theorem): For all distribution of  $X$ :

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  - A confidence interval estimation of  $\theta$  with a confidence level  $1 - \alpha$  is a random interval  $[\hat{\theta}_1; \hat{\theta}_2] = [h_1(X_1, X_2, \dots, X_n); h_2(X_1, X_2, \dots, X_n)]$  such that:

$$P(\hat{\theta}_1 \leq \theta \leq \hat{\theta}_2) = 1 - \alpha$$

## Example

Let  $X$  be the electricity bills (thousands dong) of households in a region of Vietnam (in June 2020). Observed the electricity bills of 200 households from this region and obtained the following data:

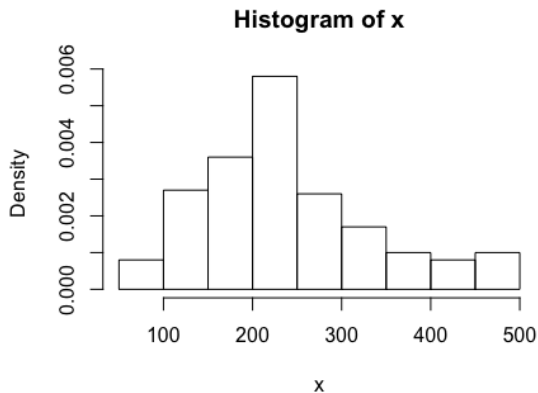
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196.65 468.75 320.50 300.50 213.05 140.60 290.00 216.95 360.50 317.95 195.55
220.50 255.60 289.00 194.55 374.25 382.05 185.55 219.10 215.60 220.00 186.75
 97.80 340.50  88.50 209.50 234.04 333.00 291.10 108.50 245.00 184.00 153.50
219.50 214.15 155.20 140.40 108.50 410.00 125.50 220.30 160.00 300.50 310.20
244.40 194.50 210.20 360.00 456.50 237.40 235.00 203.25 109.20 240.15 260.50
275.50 101.55 455.50 246.25 291.55 262.00 378.65 194.50 248.00 262.92  85.75
248.00 204.75 310.70 213.10 320.50 125.60 110.25  77.35 119.50 313.50 222.00
388.10 110.50 160.00 210.00 310.30 380.10 281.00 105.35 280.15 188.80 272.50
103.40 213.50 280.50 119.50 166.10 180.50 212.00 154.75 100.50 452.60 436.35
225.00 124.30 170.00 127.35 107.90 140.00 195.00 315.10 241.05 168.00 120.50
223.95 237.05 285.45 100.50 228.55 248.70 175.80 466.05 219.00 216.00 425.50
390.00 176.85 240.50 226.00 108.70 160.00 470.50 225.00 440.00 265.00 162.80
260.50 175.80  73.05 460.50 263.60  59.50 198.00 416.50 315.50 155.00 190.00
158.50 225.00 266.70 153.60 238.00 297.60 201.75 240.50 270.90 196.65 299.20
 70.50 125.60 100.40 240.00 240.00 224.05 194.00 247.00 325.40 102.20 166.10
361.00 430.00 240.00 250.50 470.00 157.75  98.40 236.50 230.85 317.65 200.70
165.00 350.50 319.15 275.88 203.05 234.50 220.75 180.50 436.50 403.00 460.50
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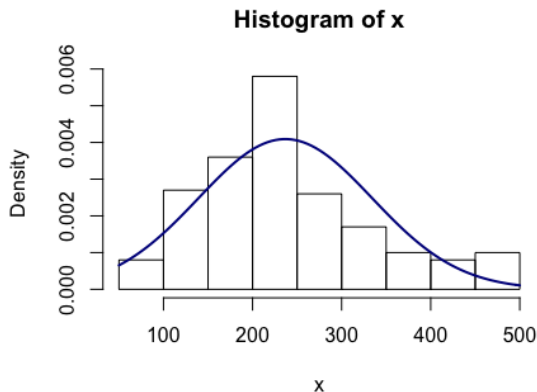
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The histogram for these data is the following:



# Example

The distribution of data can be approximated by a normal distribution:



## Example

- Modelling: We can suppose that the electricity bills of households in this region follows a normal distribution with parameter  $\theta = (\mu, \sigma^2)$  and the probability density function:

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For the given sample, the sample mean

$$\bar{x} = \frac{1}{n}(x_1 + x_2 + \dots + x_n) = 236.78$$

is also called a point estimate of  $\mu$ .

# Maximum likelihood estimation (MLE)

- Consider a random sample  $(X_1, X_2, \dots, X_n)$  taken from a population  $X$  that follows a distribution  $f(x; \theta)$  (the pdf or pmf).

# Maximum likelihood estimation (MLE)

- Consider a random sample  $(X_1, X_2, \dots, X_n)$  taken from a population  $X$  that follows a distribution  $f(x; \theta)$  (the pdf or pmf).
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- Maximum likelihood estimator: To find  $\theta$  that maximizes the likelihood function  $L(\theta)$  or  $\log L(\theta)$ :

$$\hat{\theta} = \operatorname{argmax} L(\theta) = \operatorname{argmax} \log L(\theta)$$

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$$\log L(\theta) = \sum_{i=1}^n \log f(X_i; \theta)$$

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- Step 4: Solve the equation

$$\frac{\partial \log L(\theta)}{\partial \theta} = 0$$

and let  $\hat{\theta}$  be the solution, then prove that

$$\frac{\partial^2 \log L(\theta)}{\partial \theta^2} \Big|_{\theta=\hat{\theta}} < 0$$

# Maximum likelihood estimation (MLE)

**Example 4.1:** Let  $X$  be the lifetime of a type of batteries produced by a factory and suppose that  $X$  follows an exponential distribution with a parameter  $\lambda > 0$ . Find the maximum likelihood estimator of  $\lambda$ .

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- Step 3: The log-likelihood function:

$$\log L(\lambda) = n \log \lambda - \lambda \sum_{i=1}^n X_i$$

# Maximum likelihood estimation (MLE)

- Step 4: Solve the equation

$$\frac{\partial \log L(\lambda)}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^n X_i = 0$$

we obtain the solution

$$\hat{\lambda} = \frac{n}{\sum_{i=1}^n X_i} = \frac{1}{\bar{X}}.$$

# Maximum likelihood estimation (MLE)

- Step 4: Solve the equation

$$\frac{\partial \log L(\lambda)}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^n X_i = 0$$

we obtain the solution

$$\hat{\lambda} = \frac{n}{\sum_{i=1}^n X_i} = \frac{1}{\bar{X}}.$$

- Since

$$\frac{\partial^2 \log L(\lambda)}{\partial \lambda^2} = -\frac{n}{\lambda^2} < 0, \text{ for all } \lambda > 0,$$

then the maximum likelihood estimator of  $\lambda$  is

$$\hat{\lambda} = \frac{1}{\bar{X}}.$$

# Maximum likelihood estimation (MLE) of normal distribution

- Consider a random sample  $(X_1, X_2, \dots, X_n)$  drawn from a normal population  $X$  with a mean of  $\mu$  and a variance of  $\sigma^2$ . Find the MLE of  $\theta = (\mu, \sigma^2)$ .

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- The log-likelihood function is

$$\log L(\theta) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

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- Obtain the MLE of  $\mu$  and  $\sigma^2$  as follows:

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \text{ and } \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

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# Method of moments

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- **Method of moments:** Let  $X$  be a population with a probability distribution  $f(x; \theta)$ , where  $\theta$  is an unknown parameter in  $R^r$ . The estimator of  $\theta$  by the method of moment is the solution of the following system of equations:

$$E[X^k] = \frac{1}{n} \sum_{i=1}^n X_i^k, k = 1, \dots, r.$$

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**Example 4.2:** Let  $X$  be the lifetime of a type of batteries produced by a factory and suppose that  $X$  follows an exponential distribution with a parameter  $\lambda > 0$ . Find the estimator of  $\lambda$  by the method of moments.

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- The estimator of  $\lambda$  by the method of moments is  $\hat{\lambda}_{MM} = \frac{1}{\bar{X}} = \hat{\lambda}_{MLE}$ .

# Unbiased estimator

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$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

# Confidence interval estimation

- A confidence interval estimation of  $\theta$  with a confidence level  $1 - \alpha$  is a random interval  $[\hat{\theta}_1; \hat{\theta}_2]$  such that:  $P(\hat{\theta}_1 \leq \theta \leq \hat{\theta}_2) = 1 - \alpha$ .

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- Procedure of finding a confidence interval estimation:
  - Find a point estimator  $\hat{\theta}$  of  $\theta$ .
  - Using the sampling distribution of  $\hat{\theta}$  or the central limit theorem:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0; 1)$$

to find an interval  $[\hat{\theta}_1, \hat{\theta}_2]$  such that  $P[\hat{\theta}_1 < \theta < \hat{\theta}_2] = 1 - \alpha$  (where  $\mu$  and  $\sigma$  are functions of  $\theta$ ).

## Confidence interval estimation of $\mu$

**Problem 1:** Consider a random sample  $(X_1, X_2, \dots, X_n)$  taken from a population  $X$  with a mean of  $\mu = E(X)$  and a variance of  $\sigma^2 = V(X)$ . Find a  $1 - \alpha$  confidence interval estimation of  $\mu$ .

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- Let  $Z_{\alpha/2}$  be the critical value of  $N(0; 1)$  at level  $1 - \alpha/2$ , it means that  $P(Z < Z_{\alpha/2}) = 1 - \alpha/2$ .

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- We have

$$\mathbb{P}\left(-Z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq Z_{\alpha/2}\right) = 1 - \alpha$$

$$\mathbb{P}\left(\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

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- Then a  $1 - \alpha$  confidence interval (CI) estimation of  $\mu$  is:

$$\left[\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}; \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right] = \bar{X} \mp \epsilon,$$

where  $\epsilon = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$  is called the error of CI.

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Solution:

**Case 2:** The population  $X$  is normal:  $X \sim N(\mu, \sigma^2)$ , where  $\sigma^2$  is unknown.

- A point estimator of  $\mu$  is the random sample mean  $\bar{X}$ .
- We use the T-statistic  $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ , where  $t_{n-1}$  is the Student's distribution with  $n - 1$  degrees of freedom.
- Let  $t_{n-1;\alpha/2}$  be the critical value of  $t_{n-1}$  at level  $1 - \alpha/2$ , it means that  $P(t_{n-1} < t_{n-1;\alpha/2}) = 1 - \alpha/2$ .
- We have

$$\mathbb{P}\left(-t_{n-1;\alpha/2} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{n-1;\alpha/2}\right) = 1 - \alpha$$

## Confidence interval estimation of $\mu$

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$$\mathbb{P}\left(\bar{X} - t_{n-1; \alpha/2} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{n-1; \alpha/2} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

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- The statistic  $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0; 1)$ .

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**Example 4.3:** Let  $X$  be the amount of telephone bills (USD) of customers in a city. Suppose that  $X$  follows a normal distribution  $N(\mu, \sigma^2)$ . Observed a sample of 20 customers, we obtained the following data:

31.3, 28.8, 30.8, 29.6, 32.5, 30.1, 28.6, 32.2, 30.8, 32.6,

31.8, 28.5, 29.9, 27.2, 36.0, 30.6, 29.2, 30.9, 31.0, 30.8

- Find the point estimate of  $\mu$  and  $\sigma^2$  by method of moments.
- Find the point estimate of  $\mu$  and  $\sigma^2$  by MLE method.
- Find a 90% confidence interval estimate of  $\mu$ .
- Suppose that the standard deviation  $\sigma$  is known to equal to 1.5. Find a 90% confidence interval estimate of  $\mu$ .

# Confidence interval estimation of $\mu$

**Solution of Example:** The point estimator of  $\mu$  and  $\sigma^2$  by method of moments:

- The parameter  $\theta = (\mu, \sigma^2)$  is in  $R^2$ , so the dimension  $r = 2$ .
- We have  $E(X) = \mu$  and  $E(X^2) = \mu^2 + \sigma^2$ .

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$$E[X] = \frac{1}{n} \sum_{i=1}^n X_i \text{ and } E(X^2) = \frac{1}{n} \sum_{i=1}^n X_i^2$$

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- For the given sample, the point estimate of  $\mu$  and  $\sigma^2$  by method of moments are:

$$\hat{\mu}_{MM} = \bar{x} = \frac{1}{20}(31.3 + \dots + 30.8) = 30.66$$

and

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- The point estimator of  $\mu$  and  $\sigma^2$  by the maximum likelihood estimation method are:

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- Since  $X \sim N(\mu; \sigma^2)$  where  $\sigma^2$  is unknown then we use the following statistic

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- For the given sample, we have  
 $n = 20; \bar{x} = 30.66; s^2 = \frac{n}{n-1} \hat{s}^2 = 3.604; s = \sqrt{3.604} = 1.9;$   
 $1 - \alpha = 90\%$  then  $t_{n-1; \alpha/2} = t_{19; 0.05} = 1.73$

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 $1 - \alpha = 90\%$  then  $t_{n-1; \alpha/2} = t_{19; 0.05} = 1.73$
- So the CI of  $\mu$  is

$$30.66 \pm 1.73 \frac{1.9}{\sqrt{20}} = 30.66 \pm 0.735 = [29.925; 31.395]$$

## Confidence interval estimation of $\mu$

**Solution of Example:** Find a 90% confidence interval estimate of  $\mu$  when  $\sigma = 1.5$ .

- Since  $X \sim N(\mu; \sigma^2)$  where  $\sigma$  is known to equal to 1.5 then we use the following statistic

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- For the given sample, we have  $n = 120$ ;  $\bar{x} = 30.66$ ;  $1 - \alpha = 90\%$  then  $Z_{\alpha/2} = Z_{0.05} = 1.645$

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- So the CI of  $\mu$  is

$$30.66 \pm 1.645 \frac{1.5}{\sqrt{20}} = 30.66 \pm 0.55 = [30.11; 31.21]$$

## Confidence interval estimation of $\sigma^2$

**Problem 2:** Let  $(X_1, X_2, \dots, X_n)$  be a random sample taken from a normal population  $X \sim N(\mu, \sigma^2)$ . Find a  $1 - \alpha$  confidence interval estimate of  $\sigma^2$ .

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$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2,$$

where  $\chi_{n-1}^2$  is the Chi-squared distribution with  $n - 1$  degrees of freedom.

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- A  $1 - \alpha$  confidence interval estimate of  $\sigma^2$  is

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## Confidence interval estimation of $\sigma^2$

**Example 4.4:** Let  $X$  be the amount of telephone bills (USD) of customers in a city. Suppose that  $X$  follows a normal distribution  $N(\mu, \sigma^2)$ . Observed a sample of 20 customers, we obtained the following data:

31.3, 28.8, 30.8, 29.6, 32.5, 30.1, 28.6, 32.2, 30.8, 32.6,

31.8, 28.5, 29.9, 27.2, 36.0, 30.6, 29.2, 30.9, 31.0, 30.8

Find a 90% confidence interval estimate of  $\sigma^2$ .

## Confidence interval estimation of $\sigma^2$

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where  $n = 20$ ;  $s^2 = 3.604$ ;  $1 - \alpha = 0.9$  then

$$\chi_{n-1; 1-\alpha/2}^2 = \chi_{19, 0.95}^2 = 10.12; \chi_{n-1; \alpha/2}^2 = \chi_{19, 0.05}^2 = 30.14.$$

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$$\chi_{n-1; 1-\alpha/2}^2 = \chi_{19, 0.95}^2 = 10.12; \chi_{n-1; \alpha/2}^2 = \chi_{19, 0.05}^2 = 30.14.$$

- Then a 90% confidence interval estimate of  $\sigma^2$  is

$$\left[ \frac{19 * 3.604}{30.14}; \frac{19 * 3.604}{10.12} \right] = [2.27; 6.77].$$

## Confidence interval estimation of population proportion $p$

**Problem 3:** Let  $p$  be a population proportion, for example,  $p$  is the proportion of defective items in a production line. Find a  $1 - \alpha$  confidence interval estimate of  $p$ .

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- Consider a random sample of size  $n$  from the population.

## Confidence interval estimation of population proportion $p$

**Problem 3:** Let  $p$  be a population proportion, for example,  $p$  is the proportion of defective items in a production line. Find a  $1 - \alpha$  confidence interval estimate of  $p$ .

- Consider a random sample of size  $n$  from the population.
- A point estimator of  $p$  is  $\hat{p}$ , the sample proportion (example: the proportion of defective items in a sample of  $n$  items).

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- Consider a random sample of size  $n$  from the population.
- A point estimator of  $p$  is  $\hat{p}$ , the sample proportion (example: the proportion of defective items in a sample of  $n$  items).
- By the following limit theorem

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \approx N(0, 1)$$

we obtain the following  $1 - \alpha$  confidence interval estimate of  $p$ ;

$$\hat{p} \mp \epsilon = \hat{p} \mp Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where  $\epsilon = Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$  is the error of the CI.

## Confidence interval estimation of population proportion $p$

**Example 4.5:** Let  $p$  be the proportion of defective items in a production line. Examined a random sample of 120 items from the line and there were 6 defective items. Find a 90% confidence interval estimate of  $p$ .

## Confidence interval estimation of population proportion $p$

**Example 4.5:** Let  $p$  be the proportion of defective items in a production line. Examined a random sample of 120 items from the line and there were 6 defective items. Find a 90% confidence interval estimate of  $p$ .

- A 90% confidence interval estimate of  $p$  is

$$\hat{p} \mp \epsilon = \hat{p} \mp Z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where  $n = 120$ ;  $\hat{p} = 6/120 = 0.05$ ;  $1 - \alpha = 0.9$  then  
 $Z_{\alpha/2} = Z_{0.025} = 1.96$ .

## Confidence interval estimation of population proportion $p$

**Example 4.5:** Let  $p$  be the proportion of defective items in a production line. Examined a random sample of 120 items from the line and there were 6 defective items. Find a 90% confidence interval estimate of  $p$ .

- A 90% confidence interval estimate of  $p$  is

$$\hat{p} \mp \epsilon = \hat{p} \mp Z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where  $n = 120$ ;  $\hat{p} = 6/120 = 0.05$ ;  $1 - \alpha = 0.9$  then  
 $Z_{\alpha/2} = Z_{0.025} = 1.96$ .

- So the CI of  $p$  is

$$0.05 \mp 1.96 \sqrt{\frac{0.05 * 0.95}{120}} = 5\% \mp 3.9\% = [1.1\%; 8.9\%].$$

# Confidence interval estimation

**General problem:** Observe a population  $X$  with the pdf (ou pmf)  $f(x; \theta)$ , where  $\theta$  is unknown parameter to estimate. Find a  $1 - \alpha$  confidence interval estimation of  $\theta$ .

- Consider a random sample  $(X_1, X_2, \dots, X_n)$  taken from the population  $X$ .

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- Use the sampling distribution of  $\hat{\theta}$  or a limit theorem, for example:

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$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - g_1(\theta)}{g_2(\theta)/\sqrt{n}} \approx N(0; 1)$$

- From the equation

$$\mathbb{P}\left(-Z_{\alpha/2} \leq \frac{\bar{X} - g_1(\theta)}{g_2(\theta)/\sqrt{n}} \leq Z_{\alpha/2}\right) = 1 - \alpha$$

we find an interval  $[\hat{\theta}_1; \hat{\theta}_2]$  such that  $P(\hat{\theta}_1 \leq \theta \leq \hat{\theta}_2) = 1 - \alpha$ .

# Confidence interval estimation

**Example 4.6:** Let  $X$  be the lifetime (in years) of a mechanical part. Suppose that  $X$  follows an exponential distribution with a rate parameter of  $\lambda$ .

- Construct a  $1 - \alpha$  confidence interval estimation of  $\lambda$ .
- Given the following sample:

| $X$            | $[0,1]$ | $(1, 2]$ | $(2, 3]$ | $(3, 4]$ | $(4, 5]$ | $(5, 6]$ | $(6, 7]$ |
|----------------|---------|----------|----------|----------|----------|----------|----------|
| $N^o$ of parts | 20      | 12       | 8        | 3        | 3        | 2        | 2        |

Find a 90% confidence interval estimate of  $\lambda$  for this sample.

# Confidence interval estimation

## Solution:

- Since  $X \sim \mathcal{E}(\lambda)$  then the pdf of  $X$  is  $f(x; \lambda) = \lambda e^{-\lambda x}$ , for  $x > 0$  and  $\mu = E(X) = 1/\lambda$ ;  $\sigma^2 = V(X) = 1/\lambda^2$  then  $\sigma = 1/\lambda$ .

# Confidence interval estimation

## Solution:

- Since  $X \sim \mathcal{E}(\lambda)$  then the pdf of  $X$  is  $f(x; \lambda) = \lambda e^{-\lambda x}$ , for  $x > 0$  and  $\mu = E(X) = 1/\lambda$ ;  $\sigma^2 = V(X) = 1/\lambda^2$  then  $\sigma = 1/\lambda$ .
- By the central limit theorem:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - 1/\lambda}{(1/\lambda)/\sqrt{n}} = (\bar{X}\lambda - 1)\sqrt{n} \approx N(0; 1)$$

# Confidence interval estimation

## Solution:

- Since  $X \sim \mathcal{E}(\lambda)$  then the pdf of  $X$  is  $f(x; \lambda) = \lambda e^{-\lambda x}$ , for  $x > 0$  and  $\mu = E(X) = 1/\lambda$ ;  $\sigma^2 = V(X) = 1/\lambda^2$  then  $\sigma = 1/\lambda$ .
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- From the equation

$$\mathbb{P}\left(-Z_{\alpha/2} \leq (\bar{X}\lambda - 1)\sqrt{n} \leq Z_{\alpha/2}\right) = 1 - \alpha$$

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## Solution:

- Since  $X \sim \mathcal{E}(\lambda)$  then the pdf of  $X$  is  $f(x; \lambda) = \lambda e^{-\lambda x}$ , for  $x > 0$  and  $\mu = E(X) = 1/\lambda$ ;  $\sigma^2 = V(X) = 1/\lambda^2$  then  $\sigma = 1/\lambda$ .
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- From the equation

$$\mathbb{P}\left(-Z_{\alpha/2} \leq (\bar{X}\lambda - 1)\sqrt{n} \leq Z_{\alpha/2}\right) = 1 - \alpha$$

or

$$\mathbb{P}\left(\frac{1 - \frac{Z_{\alpha/2}}{\sqrt{n}}}{\bar{X}} \leq \lambda \leq \frac{1 + \frac{Z_{\alpha/2}}{\sqrt{n}}}{\bar{X}}\right) = 1 - \alpha$$

# Confidence interval estimation

## Solution:

- Then a  $1 - \alpha$  confidence interval estimation of  $\lambda$  is

$$\left[ \frac{1 - \frac{Z_{\alpha/2}}{\sqrt{n}}}{\bar{X}}; \frac{1 + \frac{Z_{\alpha/2}}{\sqrt{n}}}{\bar{X}} \right]$$

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- For the given sample, we have  $n = 50$ ;  $1 - \alpha = 0.9$  then  
 $Z_{\alpha/2} = Z_{0.05} = 1.645$ ;  
 $\bar{x} = (20 * 0.5 + 12 * 1.5 + \dots + 2 * 6.5)/50 = 1.92$ .

# Confidence interval estimation

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- Then a  $1 - \alpha$  confidence interval estimation of  $\lambda$  is

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- So the 90% CI of  $\lambda$  is

$$\frac{1 \mp \frac{1.645}{\sqrt{50}}}{1.92} = [0.4; 0.64]$$

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## Solution:

- Then a  $1 - \alpha$  confidence interval estimation of  $\lambda$  is

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- For the given sample, we have  $n = 50$ ;  $1 - \alpha = 0.9$  then  
 $Z_{\alpha/2} = Z_{0.05} = 1.645$ ;  
 $\bar{x} = (20 * 0.5 + 12 * 1.5 + \dots + 2 * 6.5) / 50 = 1.92$ .
- So the 90% CI of  $\lambda$  is

$$\frac{1 \mp \frac{1.645}{\sqrt{50}}}{1.92} = [0.4; 0.64]$$

- We are 90% confident that the parameter  $\lambda$  is between 0.4 and 0.64.

# Confidence interval estimation

**Solution:** 2<sup>nd</sup> method

- Using the following limit theorem:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{\bar{X} - 1/\lambda}{S/\sqrt{n}} \approx N(0; 1)$$

# Confidence interval estimation

**Solution:** 2<sup>nd</sup> method

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- From the equation

$$\mathbb{P}\left(-Z_{\alpha/2} \leq \frac{\bar{X} - 1/\lambda}{S/\sqrt{n}} \leq Z_{\alpha/2}\right) = 1 - \alpha$$

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$$\mathbb{P}\left(\frac{1}{\bar{X} + Z_{\alpha/2} \frac{S}{\sqrt{n}}} \leq \lambda \leq \frac{1}{\bar{X} - Z_{\alpha/2} \frac{S}{\sqrt{n}}}\right) = 1 - \alpha$$

# Confidence interval estimation

## Solution:

- Then a  $1 - \alpha$  confidence interval estimation of  $\lambda$  is

$$\left[ \frac{1}{\bar{X} + Z_{\alpha/2} \frac{s}{\sqrt{n}}}; \frac{1}{\bar{X} - Z_{\alpha/2} \frac{s}{\sqrt{n}}} \right]$$

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- For the given sample, we have  $n = 50$ ;  $Z_{\alpha/2} = Z_{0.05} = 1.645$ ;  
 $\bar{x} = 1.92$ ;  $s^2 = (20 * 0.5^2 + \dots + 2 * 6.5^2 - 50 * 1.92^2) / 49 = 2.861$ ;  
 $s = \sqrt{2.861} = 1.69$

# Confidence interval estimation

## Solution:

- Then a  $1 - \alpha$  confidence interval estimation of  $\lambda$  is

$$\left[ \frac{1}{\bar{X} + Z_{\alpha/2} \frac{s}{\sqrt{n}}}; \frac{1}{\bar{X} - Z_{\alpha/2} \frac{s}{\sqrt{n}}} \right]$$

- For the given sample, we have  $n = 50$ ;  $Z_{\alpha/2} = Z_{0.05} = 1.645$ ;  
 $\bar{x} = 1.92$ ;  $s^2 = (20 * 0.5^2 + \dots + 2 * 6.5^2 - 50 * 1.92^2) / 49 = 2.861$ ;  
 $s = \sqrt{2.861} = 1.69$
- So the 90% CI of  $\lambda$  is

$$\frac{1}{1.92 \mp 1.645 * \frac{1.69}{\sqrt{50}}} = [0.43; 0.65]$$

# Confidence interval estimation

## Solution:

- Then a  $1 - \alpha$  confidence interval estimation of  $\lambda$  is

$$\left[ \frac{1}{\bar{X} + Z_{\alpha/2} \frac{s}{\sqrt{n}}}; \frac{1}{\bar{X} - Z_{\alpha/2} \frac{s}{\sqrt{n}}} \right]$$

- For the given sample, we have  $n = 50$ ;  $Z_{\alpha/2} = Z_{0.05} = 1.645$ ;  
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 $s = \sqrt{2.861} = 1.69$
- So the 90% CI of  $\lambda$  is

$$\frac{1}{1.92 \mp 1.645 * \frac{1.69}{\sqrt{50}}} = [0.43; 0.65]$$

- We are 90% confident that the parameter  $\lambda$  is between 0.43 and 0.65.

# Confidence interval estimation

**Example 4.7:** Let  $X$  be the number of accidents per week in a small city. Suppose that  $X$  follows a Poisson distribution with a mean parameter of  $\lambda$ .

- Find the point estimator of  $\lambda$  by the method of moment and by the MLE method.
- Construct a  $1 - \alpha$  confidence interval estimation of  $\lambda$ .
- Given the following sample:

| $X$            | 0 | 1  | 2  | 3  | 4 |
|----------------|---|----|----|----|---|
| $N^o$ of weeks | 7 | 15 | 10 | 12 | 6 |

Find a 90% confidence interval estimate of  $\lambda$  for this sample.

# Confidence interval estimation

**Solution:** The point estimator of  $\lambda$  by the method of moment:

- The parameter  $\lambda \in R_+^*$  then the dimension of parameter space is  $r = 1$ .

# Confidence interval estimation

**Solution:** The point estimator of  $\lambda$  by the method of moment:

- The parameter  $\lambda \in R_+^*$  then the dimension of parameter space is  $r = 1$ .
- The first moment of  $X$  is  $E(X) = \lambda$ .

# Confidence interval estimation

**Solution:** The point estimator of  $\lambda$  by the method of moment:

- The parameter  $\lambda \in R_+^*$  then the dimension of parameter space is  $r = 1$ .
- The first moment of  $X$  is  $E(X) = \lambda$ .
- The first sample moment of  $X$  is  $\frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$ .

# Confidence interval estimation

**Solution:** The point estimator of  $\lambda$  by the method of moment:

- The parameter  $\lambda \in R_+^*$  then the dimension of parameter space is  $r = 1$ .
- The first moment of  $X$  is  $E(X) = \lambda$ .
- The first sample moment of  $X$  is  $\frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$ .
- We solve the following equation:

$$E(X) = \frac{1}{n} \sum_{i=1}^n X_i \text{ or } \lambda = \bar{X}$$

So the point estimator of  $\lambda$  by the method of moment is  $\hat{\lambda}_{MM} = \bar{X}$ .

# Confidence interval estimation

**Solution:** The point estimator of  $\lambda$  by the MLE method.:

- The pmf of  $X$  is

$$f(x; \lambda) = e^{-\lambda} \frac{\lambda^x}{x!}, x = 0, 1, 2, \dots$$

# Confidence interval estimation

**Solution:** The point estimator of  $\lambda$  by the MLE method.:

- The pmf of  $X$  is

$$f(x; \lambda) = e^{-\lambda} \frac{\lambda^x}{x!}, x = 0, 1, 2, \dots$$

- The likelihood function of  $\lambda$  is

$$L(\lambda) = \prod_{i=1}^n f(X_i; \lambda) = \prod_{i=1}^n e^{-\lambda} \frac{\lambda^{X_i}}{X_i!} = e^{-n\lambda} \frac{\lambda^{\sum_{i=1}^n X_i}}{\prod_{i=1}^n X_i!}$$

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- The log-likelihood function of  $\lambda$  is

$$\log L(\lambda) = -n\lambda + \sum_{i=1}^n X_i \log(\lambda) - \log\left(\prod_{i=1}^n X_i!\right)$$

# Confidence interval estimation

**Solution:** The point estimator of  $\lambda$  by the MLE method:

- Solve the following equation:

$$\frac{\partial \log L(\lambda)}{\partial \lambda} = -n + \frac{\sum_{i=1}^n X_i}{\lambda} = 0$$

we have  $\lambda = \bar{X}$ .

# Confidence interval estimation

**Solution:** The point estimator of  $\lambda$  by the MLE method:

- Solve the following equation:

$$\frac{\partial \log L(\lambda)}{\partial \lambda} = -n + \frac{\sum_{i=1}^n X_i}{\lambda} = 0$$

we have  $\lambda = \bar{X}$ .

- Since

$$\frac{\partial^2 \log L(\lambda)}{\partial \lambda^2} = -\frac{\sum_{i=1}^n X_i}{\lambda^2} < 0$$

then the likelihood function attains maximum at  $\lambda = \bar{X}$ . So  $\hat{\lambda}_{MLE} = \bar{X}$ .

# Confidence interval estimation

**Solution:** Construct a  $1 - \alpha$  confidence interval estimation of  $\lambda$ :

- Since  $X \sim P(\lambda)$  then  $\mu = E(X) = \lambda$ ;  $\sigma^2 = V(X) = \lambda$ ;  $\sigma = \sqrt{\lambda}$ .

# Confidence interval estimation

**Solution:** Construct a  $1 - \alpha$  confidence interval estimation of  $\lambda$ :

- Since  $X \sim P(\lambda)$  then  $\mu = E(X) = \lambda$ ;  $\sigma^2 = V(X) = \lambda$ ;  $\sigma = \sqrt{\lambda}$ .
- By the central limit theorem:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - \lambda}{\sqrt{\lambda}} \sqrt{n} \approx N(0; 1)$$

# Confidence interval estimation

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- From the equation

$$\mathbb{P}\left(\frac{|\bar{X} - \lambda|}{\sqrt{\lambda}} \sqrt{n} \leq Z_{\alpha/2}\right) = 1 - \alpha$$

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- From the equation

$$\mathbb{P}\left(\frac{|\bar{X} - \lambda|}{\sqrt{\lambda}} \sqrt{n} \leq Z_{\alpha/2}\right) = 1 - \alpha$$

or

$$\mathbb{P}\left(n(\bar{X} - \lambda)^2 \leq \lambda Z_{\alpha/2}^2\right) = 1 - \alpha$$

# Confidence interval estimation

**Solution:** Construct a  $1 - \alpha$  confidence interval estimation of  $\lambda$ :

• or

$$\mathbb{P}\left(n\lambda^2 - (2n\bar{X} + Z_{\alpha/2}^2)\lambda + n\bar{X}^2\right) = 1 - \alpha$$

# Confidence interval estimation

**Solution:** Construct a  $1 - \alpha$  confidence interval estimation of  $\lambda$ :

• or

$$\mathbb{P}\left(n\lambda^2 - (2n\bar{X} + Z_{\alpha/2}^2)\lambda + n\bar{X}^2\right) = 1 - \alpha$$

• or

$$\mathbb{P}\left(\lambda \in \frac{2n\bar{X} + Z_{\alpha/2}^2 \mp Z_{\alpha/2}\sqrt{4n\bar{X} + Z_{\alpha/2}^2}}{2n}\right) = 1 - \alpha$$

# Confidence interval estimation

**Solution:** Construct a  $1 - \alpha$  confidence interval estimation of  $\lambda$ :

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$$\mathbb{P}\left(n\lambda^2 - (2n\bar{X} + Z_{\alpha/2}^2)\lambda + n\bar{X}^2\right) = 1 - \alpha$$

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$$\mathbb{P}\left(\lambda \in \frac{2n\bar{X} + Z_{\alpha/2}^2 \mp Z_{\alpha/2}\sqrt{4n\bar{X} + Z_{\alpha/2}^2}}{2n}\right) = 1 - \alpha$$

- Then a  $1 - \alpha$  confidence interval estimation of  $\lambda$  is

$$\frac{2n\bar{X} + Z_{\alpha/2}^2 \mp Z_{\alpha/2}\sqrt{4n\bar{X} + Z_{\alpha/2}^2}}{2n}$$

# Confidence interval estimation

## Solution:

- For the given sample, we have  $n = 50$ ;  $1 - \alpha = 0.9$  then  
 $Z_{\alpha/2} = Z_{0.05} = 1.645$ ;  
 $\bar{x} = (7 * 0 + 15 * 1 + 10 * 2 + 21 * 3 + 6 * 4) / 50 = 1.9$ .

# Confidence interval estimation

## Solution:

- For the given sample, we have  $n = 50$ ;  $1 - \alpha = 0.9$  then

$$Z_{\alpha/2} = Z_{0.05} = 1.645;$$

$$\bar{x} = (7 * 0 + 15 * 1 + 10 * 2 + 21 * 3 + 6 * 4) / 50 = 1.9.$$

- So the 90% CI of  $\lambda$  is

$$\frac{2 * 50 * 1.9 \pm 1.645 \sqrt{4 * 50 * 1.9 + 1.645^2}}{2 * 50} = [1.61; 2.25]$$

# Confidence interval estimation

## Solution:

- For the given sample, we have  $n = 50$ ;  $1 - \alpha = 0.9$  then

$$Z_{\alpha/2} = Z_{0.05} = 1.645;$$

$$\bar{x} = (7 * 0 + 15 * 1 + 10 * 2 + 21 * 3 + 6 * 4) / 50 = 1.9.$$

- So the 90% CI of  $\lambda$  is

$$\frac{2 * 50 * 1.9 \pm 1.645^2 \mp 1.645 \sqrt{4 * 50 * 1.9 + 1.645^2}}{2 * 50} = [1.61; 2.25]$$

- We are 90% confident that the parameter  $\lambda$  is between 1.61 and 2.25.

# Confidence interval estimation

**Solution:** 2<sup>nd</sup> method.

- By the following limit theorem:

$$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{\bar{X} - \lambda}{S} \sqrt{n} \approx N(0; 1)$$

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- From the equation

$$\mathbb{P}\left(\frac{|\bar{X} - \lambda|}{S} \sqrt{n} \leq Z_{\alpha/2}\right) = 1 - \alpha$$

or

$$\mathbb{P}\left(\bar{X} - Z_{\alpha/2} \frac{S}{\sqrt{n}} \leq \lambda \leq \bar{X} + Z_{\alpha/2} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

# Confidence interval estimation

## Solution:

- Then a  $1 - \alpha$  confidence interval estimation of  $\lambda$  is

$$\bar{X} \mp Z_{\alpha/2} \frac{S}{\sqrt{n}}$$

# Confidence interval estimation

## Solution:

- Then a  $1 - \alpha$  confidence interval estimation of  $\lambda$  is

$$\bar{X} \mp Z_{\alpha/2} \frac{S}{\sqrt{n}}$$

- For the given sample, we have  $n = 50$ ;  $Z_{\alpha/2} = Z_{0.05} = 1.645$ ;  
 $\bar{x} = 1.9$ ;  $s^2 = (7 * 0^2 + 15 * 1^2 + 10 * 2^2 + 21 * 3^2 + 6 * 4^2 - 50 * 1.9^2) = 1.602$ ;  $s = \sqrt{1.602} = 1.27$ .

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- So the 90% CI of  $\lambda$  is

$$1.9 \mp 1.645 \frac{1.27}{\sqrt{50}} = [1.6; 2.2]$$

- We are 90% confident that the parameter  $\lambda$  is between 1.6 and 2.2.